ABSTRACT

Agriculture faces challenges such as crop diseases, pests, nutrient deficiencies, improper irrigation and changing weather conditions. Farmers often depend on manual inspection, making early detection difficult and increasing the risk of crop losses, water wastage and delayed decisions.

The proposed **IoT-Enabled AI-Based Smart Farming Rover** uses cameras and sensors to collect crop and environmental data. AI/ML models analyze crop health, diseases, pests and nutrient deficiencies, while sensor data helps assess water stress and irrigation needs. Edge computing enables faster field-level analysis, with alerts and recommendations provided through a mobile application.

INTRODUCTION

Agriculture is a primary source of livelihood and food security, and its productivity depends on timely awareness of crop and field conditions [11]. Crops are continuously exposed to diseases, pest infestations, nutrient deficiencies and water stress, as well as environmental risks such as erratic rainfall, drought, flooding, heat stress and fire. Conventional monitoring relies largely on periodic manual inspection, which is labour-intensive, difficult to scale across large or remote fields and may delay the identification of emerging problems. The Internet of Things (IoT) enables continuous, low-cost sensing of soil and environmental parameters; mobile robotic platforms can carry sensors and cameras directly into the field; and Artificial Intelligence (AI) with computer vision can help interpret field images and sensor data. Integrating these technologies into a single field-deployable rover provides a practical means of delivering real-time field information and decision support to farmers.

OVERVIEW

The proposed system is an integrated **IoT + AI + Robotics** agricultural monitoring system built around a mobile farming rover. The rover carries environmental sensors (temperature, humidity, soil moisture, rain and flame), ultrasonic sensors for obstacle detection and an ESP-CAM module for field imagery. ESP32 controllers acquire sensor readings, drive the motors and transmit data over Wi-Fi using WebSocket/HTTP to a Node.js/Express backend. The backend stores data in MongoDB, combines it with external weather and environmental data from public APIs, and applies Python-based AI/computer vision analysis. The resulting insights and alerts are presented through a web dashboard and a mobile-friendly farmer interface, enabling near real-time monitoring of field conditions.

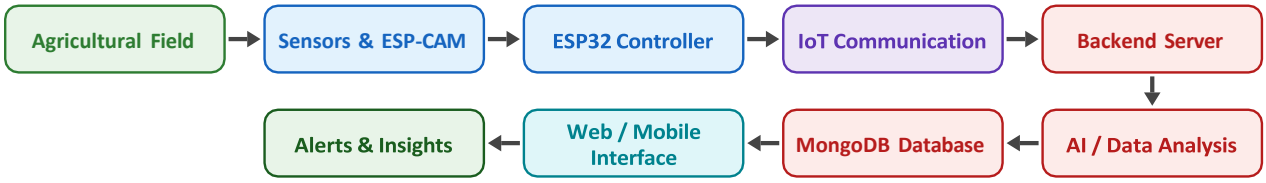


Figure 1 Overall data flow of the proposed system

LITERATURE SURVEY

1. **IoT-Based Smart Agriculture** – Wireless sensor networks and IoT platforms are widely used to monitor soil, climate and crop parameters with remote data access [1], [2]. *Gap:* most deployments use fixed sensor nodes covering limited points and provide no visual crop information.
2. **AI / Computer Vision for Crop Monitoring** – Deep convolutional neural networks have shown strong results in identifying plant diseases from leaf images [3], [4]. *Gap:* many models are trained on controlled datasets and are seldom linked with live field sensing.
3. **Sensor-Based Irrigation and Soil Monitoring** – Soil-moisture and weather sensing are used to schedule irrigation and reduce water wastage [5], [6]. *Gap:* decisions often depend on a single parameter without combining weather and crop condition.
4. **Agricultural Robotics** – Field robots have been developed for inspection, spraying and harvesting [7]. *Gap:* many platforms are costly and target a single operation rather than integrated monitoring.
5. **Edge / Embedded Intelligence** – Edge computing processes data close to its source to reduce latency and bandwidth [8]. *Gap:* low-power microcontrollers limit on-device analysis, requiring a balanced edge–server design.
6. **Environmental Monitoring and Data-Driven Farming** – Big-data and deep-learning approaches combine weather, satellite and field data for farm decision support [9], [10]. *Gap:* such data are rarely combined with ground-level rover observations in low-cost systems.

MOTIVATION

Crop and environmental problems often develop gradually and are easier to manage when noticed early, yet continuous manual observation of every part of a field is impractical. Repetitive inspection is physically demanding, some areas may be difficult or unsafe to access during fire or adverse weather, and irrigation decisions are frequently made without reliable soil-moisture or weather information. IoT sensing enables continuous, low-cost measurement of field parameters; AI-assisted image and data analysis can help identify possible crop and environmental risks; and a mobile rover provides remote visibility of the field. Integrating embedded hardware with backend, analytics and user-facing software is essential to convert raw readings into timely and understandable information for farmers.

PROBLEM STATEMENT

Agricultural productivity is affected by crop diseases, pest infestations, nutrient deficiencies, water stress, erratic rainfall, droughts, floods, heat waves and other environmental risks. Conventional monitoring depends largely on periodic manual inspection, which may delay the identification of emerging problems and is difficult to scale across large or remote fields. There is therefore a need for a field-deployable intelligent monitoring system capable of continuously collecting and analysing crop and environmental information. This project develops an IoT-enabled, AI-based agricultural rover that integrates embedded sensors, camera-based monitoring, robotics, wireless communication, AI/computer vision and software analytics to provide real-time field information and actionable decision support.

INPUT: Environmental sensor readings, ESP-CAM field images and external weather/environmental data.
PROCESS: Embedded data acquisition, wireless transmission, backend processing, storage and AI/computer-vision-assisted analysis.

OUTPUT: Real-time field status, alerts and decision-support information on web and mobile interfaces.

OBJECTIVES

- To develop a mobile agricultural rover platform for field monitoring.
- To integrate environmental and agricultural sensors with ESP32-based embedded controllers.
- To implement IoT-based wireless communication between the rover and the software system.
- To integrate ESP-CAM based camera observation of crops and field conditions.
- To apply AI/computer vision techniques to support analysis of crop and field conditions.
- To develop backend services and a MongoDB database for managing field data.
- To develop a web/mobile interface for real-time monitoring, alerts and decision support.
- To test and evaluate the integrated hardware–software prototype.

SCOPE OF THE PROJECT

- **Field Monitoring:** rover-based field inspection.
- **Environmental Sensing:** temperature, humidity, soil moisture, rain and flame.
- **IoT Communication:** Wi-Fi based WebSocket/HTTP data transfer.
- **Rover Operation:** motor-driven movement with obstacle detection.
- **Computer Vision:** camera-based crop and field observation.
- **AI-Assisted Analysis:** support in identifying crop and field risks.
- **Irrigation Support:** soil-moisture and weather-based guidance.
- **Risk Monitoring:** rainfall, heat and fire related alerts.
- **Remote Monitoring:** live data on web/mobile interfaces.
- **Historical Analysis:** stored field data for trend review.

Future scope: autonomous navigation, GPS-based field mapping, automated irrigation and advanced predictive analytics are planned enhancements and are not part of the current prototype.

METHODOLOGY

- **Field Data Acquisition:** Sensors and the camera collect environmental readings and field images.
- **Embedded Data Acquisition:** ESP32 controllers read sensor values and manage the rover hardware.
- **Rover Mobility:** BO DC motors driven by TB6612FNG drivers move the rover; ultrasonic sensors detect obstacles.
- **Image Acquisition:** The ESP-CAM, positioned using a servo motor, captures crop and field imagery.
- **AI / Computer Vision Analysis:** Image and sensor information is analysed using Python and OpenCV to help identify agricultural conditions and risks.
- **IoT Communication:** Sensor data, images and rover status are sent over Wi-Fi (WebSocket/HTTP).
- **Backend Processing:** Node.js/Express services receive, process and organise incoming data along with external weather data.
- **Data Storage:** MongoDB stores sensor readings, image references and alert history.
- **Risk Analysis:** Analysis components combine sensor, image and environmental data to generate alerts and irrigation recommendations.
- **Farmer Interface:** Results are presented through the web dashboard and mobile interface.

HARDWARE COMPONENTS

Component	Purpose	Role
ESP32 Dev Modules (×2) [12]	Sensor handling, motor control, IoT communication	Control & IoT
ESP-CAM Module	Captures and streams field images over Wi-Fi	Vision
DHT11 Sensor	Measures air temperature and humidity	Environmental Sensing
Soil Moisture Sensor	Measures soil moisture level	Environmental Sensing
Rain Sensor	Detects rainfall	Environmental Sensing
5-Channel Flame Sensor	Detects fire/flame in multiple directions	Hazard Detection
Ultrasonic Sensors (×4)	Obstacle detection and distance measurement	Navigation Safety
Servo Motors	Camera and sensor positioning	Actuation
BO DC Motors (×4)	Drive the rover wheels	Mobility
TB6612FNG Motor Drivers (×2)	Control motor direction and speed	Mobility
Buzzer	Audible alerts for fire, obstacles or hazards	Alert System
Rover Chassis & Wheels	Mounts all components and enables movement	Mechanical Structure
Power Subsystem	Supplies power to controllers, sensors and motors	Power

SOFTWARE / TECHNOLOGIES

Category	Technology	Purpose
Embedded Firmware	Embedded C/C++ (ESP32)	Sensor acquisition and motor control
Frontend	React, Tailwind CSS	Web dashboard and mobile interface
Backend	Node.js, Express.js	Data ingestion, APIs and processing
Database	MongoDB	Storage of sensor, image and alert data
Communication	Wi-Fi, WebSocket / HTTP	Real-time data transfer
AI / Computer Vision	Python, OpenCV [13], AI/ML	Image and data analysis
External Data	Environmental / public APIs	Weather, rainfall and environmental information

SYSTEM ARCHITECTURE OF THE PROPOSED SMART FARMING ROVER

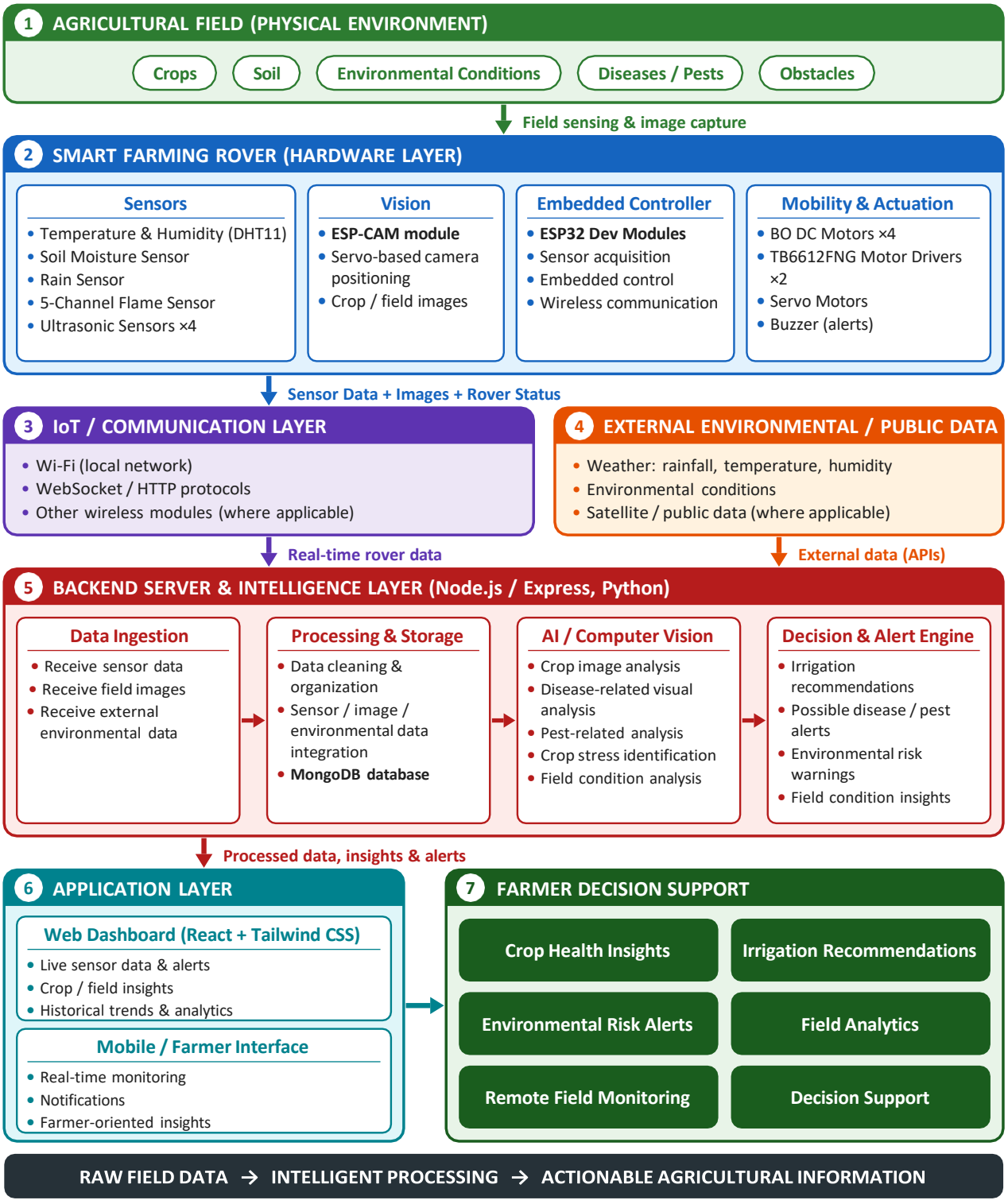


Figure 2 System Architecture of the Proposed Smart Farming Rover

The proposed architecture integrates field-level IoT sensing, rover-based data acquisition, wireless communication, AI-assisted analysis, database services and farmer-facing interfaces. Sensor and image data collected by the rover are transmitted to the software infrastructure, where they are processed and combined with relevant environmental information to generate real-time insights, alerts and decision-support information.

REAL-WORLD APPLICATION

The system can assist farmers and field staff in practical agricultural settings, including:

- Agricultural field monitoring
- Remote field inspection
- Soil moisture monitoring
- Irrigation decision support
- Crop health observation
- Environmental risk awareness
- Early identification of possible crop problems
- Reduced repetitive manual inspection
- Historical field monitoring
- Support for precision agriculture

EXPECTED OUTCOMES

- Working agricultural rover prototype
- Real-time environmental sensing
- IoT-based data communication
- Camera-based field monitoring
- AI-assisted crop and field analysis
- Backend data management (MongoDB)
- Web/mobile monitoring interface
- Alerts and decision-support functionality
- Integrated hardware–software architecture

FUTURE SCOPE

The following are proposed future enhancements beyond the current prototype:

- Autonomous navigation and path planning
- GPS-based field mapping
- Advanced disease/pest detection
- More capable edge AI on the rover
- Automated irrigation control
- Cloud-based analytics
- Digital twin of the farm
- Drone integration
- Multi-rover deployment
- Advanced predictive agricultural analytics
- Multilingual farmer assistance

REFERENCES

[1] A. Tzounis, N. Katsoulas, T. Bartzanas and C. Kittas, "Internet of Things in agriculture, recent advances and future challenges," *Biosystems Engineering*, vol. 164, pp. 31–48, 2017.

[2] O. Elijah, T. A. Rahman, I. Orikumhi, C. Y. Leow and M. N. Hindia, "An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges," *IEEE Internet of Things Journal*, vol. 5, no. 5, pp. 3758–3773, 2018.

[3] S. P. Mohanty, D. P. Hughes and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, Art. 1419, 2016.

[4] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.

[5] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour and E.-H. M. Aggoune, "Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk," *IEEE Access*, vol. 7, pp. 129551–129583, 2019.

[6] K. G. Liakos, P. Busato, D. Moshou, S. Pearson and D. Bochtis, "Machine learning in agriculture: A review," *Sensors*, vol. 18, no. 8, Art. 2674, 2018.

[7] A. Bechar and C. Vigneault, "Agricultural robots for field operations: Concepts and components," *Biosystems Engineering*, vol. 149, pp. 94–111, 2016.

[8] W. Shi, J. Cao, Q. Zhang, Y. Li and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.

[9] S. Wolfert, L. Ge, C. Verdouw and M.-J. Bogaardt, "Big data in smart farming – A review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017.

[10] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[11] FAO, *The Future of Food and Agriculture – Trends and Challenges*. Rome: Food and Agriculture Organization of the United Nations, 2017.

[12] Espressif Systems, *ESP32 Technical Reference Manual*. [Online]. Available: https://www.espressif.com

[13] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.